Mathematics-I (MAL1010)

Dr. Subhash Bhagat and Dr. Md Abu Talhamainuddin Ansary

Department of Mathematics
Indian Institute of Technology Jodhpur

July 31, 2025

Instructor(s)

Role	Name	Email
Course Coordinator	Dr. Subhash Bhagat	sbhagat@iitj.ac.in
Co-Instructor	Dr. Md Abu T. Ansary	mdabutalha@iitj.ac.in

Venue for the Lectures

Section A	LHB 110
Section B	LHB 308

Schedule Table

Days	**Monday**	**Tuesday**	**Thursday**
Time	**4:00–4:50 pm**	**8:00–8:50 am**	**8:00–8:50 am**

Tutorial Schedule

Days	Wednesday	Thursday
Time	5:00–5:50 pm	5:00–5:50 pm

Office Timings during the week:

Office hour meeting: Thursday 4-5 pm, Department of Mathematics

Contact Details of Teaching Assistants (TAs)

Please contact the TAs for any query related to the course or attendance.

Name	**Email**
Ankit Chauhan	chauhan.16@iitj.ac.in
Ajay Kumar	p22ma004@iitj.ac.in
Barsha Shaw	p22ma202@iitj.ac.in
Anjali Moyal	p24ma0005@iitj.ac.in
Mrityunjoy Saha	saha.8@iitj.ac.in
Shivam Kumar	p24ma0006@iitj.ac.in
Swaroop Dan Charan	p24ma0002@iitj.ac.in
Shubham Kumar Jain	p22ma208@iitj.ac.in
Arijit Saha	p23ma0005@iitj.ac.in
Geetanjli	p23ma0007@iitj.ac.in
Harshit Jain	p23ma0008@iitj.ac.in
Ashutosh Dixit	p23ma0014@iitj.ac.in
Nitish Kumar	p23ma0010@iitj.ac.in
Prakhar Shukla	p23ma0011@iitj.ac.in

Contact Details of Teaching Assistants (TAs) – contd.

(continued)

Name	**Email**
Gaurav Kumar	P24MA0001@iitj.ac.in
Ishita Singh	m24mac001@iitj.ac.in
Shashank Mishra	m24mac011@iitj.ac.in
Harsh Varashney	m23ma2006@iitj.ac.in
Sidharth Choudhary	M23ma2009@iitj.ac.in
Amit Kumar	m22ma202@iitj.ac.in
Ankit Dalal	m22ma203@iitj.ac.in

Content of the Course

Title	Mathematics I	Number	MAL1010
Department	*Mathematics*	*L-T-P [C]*	*3-1-0 [4]*
Offered for	*B.Tech. 1st yr*	*Type*	*Compulsory*
Prerequisite	*None*		

Objectives:

1. Train the undergraduate students towards basic understanding of Mathematics.
2. Provide student with sufficient knowledge in calculus, which can be used by the students in their respective fields.
3. Enable students to develop a working knowledge of central ideas of single and multivariable calculus.

Learning Outcomes

1. Impart knowledge of basic concepts like Limit, Continuity, Differentiability, Maxima and Minima, Riemann Integration and their applications.
2. Gain strong understanding of Integration in higher dimension and Vector Calculus.
3. Improve their ability to think critically, to analyze a problem and solve it using a wide array of tools.

Course Content

Module 1 [10 Lectures]:

- Properties of Real number system.
- Sequences and series of real numbers
- Review of limit, Continuity and differentiability of functions
- Rolle's theorem, Mean value theorems and Taylor's theorem
- Maxima, minima and curve tracing.

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Course Content

Module 2 [7 Lectures]:

- Riemann integral, Fundamental theorem of calculus
- Application to length, area, volume, surface area of revolution.

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Course Content

Module 3 [6 Lectures]:

- Functions of Several Variables
- Sequences, Limit, and continuity of functions of several variables
- Partial derivatives, Chain rule, Gradient, Directional derivative and differentiability
- Taylor's formula for functions from $\mathbb{R}^2$ to $\mathbb{R}$
- Tangent planes and normal, Maxima, minima, saddle points, Lagrange multipliers
- Total derivative of Functions from $\mathbb{R}^n$ to $\mathbb{R}^m$, Jacobian matrix, sufficient condition for differentiability.

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Course Content

Module 4 [6 Lectures]:

- Double and triple integrals
- Change of variables
- Vector fields, Gradient, Curl and Divergence, Curves

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Module 5 [10 Lectures]:

- Line integrals and their applications
- Green's theorem and applications, Surfaces, Surface area, Surface integrals
- Divergence theorem, Stokes' theorem and applications.

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Text Books:

1. Thomas, G. B. and Finney, R. L., (1992), Calculus and Analytic Geometry, 9th Edition, Addison Wesley Publishing Company
2. Ghorpade, S. R. and Limaye, B. V., (2006), A Course in Calculus and Real Analysis, Springer Verlag
3. Ghorpade, S. R. and Limaye, B. V., (2009), A Course in Multivariable Calculus and Analysis, Springer Verlag

Reference Books: 1. Thomas Jr., G.B., Weir, M.D. and Hass, J.R., (2015), Thomas' Calculus, Pearson Education. 2. Ayers, F. and Mendelson, E. (2012) Schaum's Outline of Calculus, McGraw Hill. Online Course Material: 1. Dutta, J. (IIT Kanpur) Calculus of One Real Variable, https://nptel.ac.in/courses/109104124/2

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Online Course Material:

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Evaluation policy for the course

Components	Weightage (%)	Date and Timings	Remarks
Assignments	10%	Will be mentioned separately on google classroom.	Number of assignments is 4.
Quizzes	20%	Will be announced in the class	No make-up will be arranged. Best two will be considered out of three.
Minor	25%	**17-09-25 to 20-09-25**	Written Examination
Major	45%	**20-11-25 to 26-11-25**	Closed Books, Written examinations, Scientific calculators are permitted
Projects	NA	NA	NA

Guidelines for Examination

1. As per the Institute's rules and regulations.

Attendance Policy:

1. As per institute norms, a student is expected to have full attendance in each course unless the student takes a leave of absence for valid medical or bonafide reasons. In any case, the attendance of a student should not fall below 75% in a course. The attendance records must be made available weekly to all students enrolled in a course by the instructors/TAs.

2. If a student remains absent from a class without sanctioned leave for more than two weeks, then the course instructor may recommend the de-registration of the student from the course.

3. If a student is found to be absent from all academic activities for four or more than four weeks (not necessarily contiguous) in a semester, with or without sanctioned leave, then his/her registration will result in automatic semester withdrawal for all the courses.

4. If a student's attendance falls below 75% without any valid reason, the instructors can recommend de-registration from the course to the SUGC/SPGC.

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Any other relevant information:

For any queries, you can visit our office during the mentioned office hour. Moreover, you can also post your doubts and queries on google classroom: https://classroom.google.com/c/NzkwNDAwMzg2Njg3?cjc=nmmwljzw

Table of Lecture-wise topics covered:

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Lecture	Following Topics to be Covered
Lecture 1	Introduction of the subject in general
Lecture 2	Properties of Real number, Interval: Definition and Types , Basic Inequalities
Lecture 3	Sets and their properties , Relation and their types and Functions: Definitions and basic examples, Properties of function and Intermediate value Property, Vertical line test, piecewise function, Increasing and decreasing function, Rational function, inverse function
Lecture 4	Introduction to Sequence , limit of a sequence, Cauchy first and second theorem on limits and cesaro's theorem
Lecture 5	Limit : Ɛ-δ Definition, Uniqueness Property of limit of a function, Limit laws and Sandwich theorem
Lecture 6	Continuity: Ɛ-δ Definition, properties of continuous function, Intermediate value theorem for continuity and Laws
Lecture 7	Monotone sequence and their properties, Sub sequences, limit inferior, limit superior, Bolzano- Weierstrass theorem in R and Cauchy criterion in R

Table of Lecture-wise topics covered:

Lecture 8	Derivatives : Meaning and definition, Algebra of derivatives, Properties of differentiable functions, Inverse function theorem
Lecture 9	Mean value theorems: Rolle's and Taylor's theorem
Lecture 10	Monotonicity, Convexity and Concavity, point of inflection
Lecture 11	Riemann Integration : Definition and Example
Lecture 12	Integrable function
Lecture 13	Riemann sum and Riemann integrals over bounded sets
Lecture 14	Fundamental theorem of Calculus
Lecture 15	Volume of solid of revolution by Slicing Method
Lecture 16	Volume by Cylindrical Shells and Surface area
Lecture 17	Multivariable Functions, Function in R^2R^2, R^nR^n
Lecture 18	ε-δ Definition of Limit & Continuity
Lecture 19	Partial Derivatives
Lecture 20	Differentiability in R^2R^2
Lecture 21	Differentiation Contd.
Lecture 22	Chain Rule
Lecture 23	Directional Derivatives
Lecture 24	Directional Derivatives & Gradient
Lecture 25	Taylor's theorem for multivariable functions

Table of Lecture-wise topics covered:

Lecture 26	Tangent planes and Normal
Lecture 27	Maxima-Minima and saddle point for multivariable functions
Lecture 28	Lagrange's multiplier
Lecture 29	Total derivative of multivariable functions, Jacobian matrix
Lecture 30	Multivariable integration
Lecture 31	Fubini's theorem and applications
Lecture 32	Change of variable in integration and application: area and volume
Lecture 33	Vector field, gradient, curl and divergence
Lecture 34	Line integral and its applications
Lecture 35	Green's theorem and its application
Lecture 36	Surface area and its application
Lecture 37	Surface integral and its application
Lecture 38	Gauss divergence theorem and its application
Lecture 39	Stokes theorem and its application

- Email: sbhagat@iitj.ac.in and mdabutalha@iitj.ac.in
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Thank You!